

Individual Project

Weekly Submission

Week 1 (27th January – 30th January): Problem Statement, Dataset Selection & Preprocessing

- Choose domain (Education / Social Media / Healthcare / etc.) and finalize dataset source.
- Define a clear problem statement and objectives.
- Understand dataset attributes and size.
- Perform data cleaning: handle missing values, duplicates, inconsistencies.
- Save the preprocessed dataset for Hadoop use.
- Show: dataset link, problem statement, preprocessing steps, and sample output.

Week 2 (2nd February – 6th February): HDFS

- Create HDFS directory structure for the project.
- Upload the preprocessed dataset into HDFS.
- Verify upload using HDFS commands.
- Explain block size, replication factor, and storage structure.
- Show: HDFS commands and successful upload screenshots.

Week 3 (9th February – 13th February): MapReduce

- Design at least one MapReduce problem (aggregation / filtering / counting / analysis).
- Write and execute MapReduce program.
- Interpret the output and explain results.
- Show: code, execution, and output file in HDFS.

Week 4 (16th February – 20th February): Hive

- Create Hive database and tables from HDFS data.
- Load data and perform analytical queries.
- Generate meaningful insights using SQL queries.
- Show: table schema, queries, and results.

Week 5 (23rd February – 27th February): Pig & Hue

- Write Pig scripts for data analysis.
- Use Hue interface to explore and query data.
- Compare Pig and Hive results (if applicable).
- Show: Pig scripts, Hue screenshots, and insights.

Week 6 (2nd March – 6th March): HBase & Sqoop

- Store selected data into HBase tables.
- Demonstrate Sqoop import/export between RDBMS and HDFS/Hive (if applicable).
- Explain use cases of HBase and Sqoop in the project.
- Show: table creation, data insertion, and Sqoop commands.

Week 7 (9th March – 13th March): Data Visualization & Final Insights

- Create visualizations from Hadoop outputs.
- Summarize key findings and insights.
- Discuss business/social impact, limitations, and future scope.
- Prepare final report and presentation for submission.

Final Submission (13th March 2026): Project Report, Code Folder, Presentation, and Weekly Progress Log